MIL-C-915/35A
1 August 1972
SUPERSEDING 1/
MIL-C-915/35(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPES TCJU AND TCTU

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (watertight)

(A)
First - One iron and one constantan conductor in type TCJU, for type J thermocouple in accordance with American National Standard-C96.1; one copper and one constantan in type TCTU, for type T thermocouple in accordance with American National Standard-C96.1; (see details for size). All conductors uncoated.
Second - Extruded silicone rubber insulation (see details for diameter).
Third - Glass braid.
(A) Fourth - Braid covering. Printed identification as follows: Type TCJU shall have the legend "8 (GREY)" printed on the iron conductor, and the legend "3 (RED)" printed on the constantan conductor. Type TCTU shall have the legend "6 (BLUE)" printed on the copper conductor, and the legend "3 (RED)" printed on the constantan conductor (see details for diameter).
Fifth - The two conductors shall be cabled together with a lay of 16 to 24 times the pitch diameter. Non-fibrous fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder/barrier.
(A) Seventh - Standard thermoplastic jacket (gray). Nominal thickness 0.050 inch. Cable surface marking required.

(A) Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(a) The type designation shall change to TCJA or TCTA,, whichever is applicable.
(b) Standard thermoplastic jacket (black or gray) shall be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage.

(A)

Details

Type and size	Conductor			Overall diameter		(A) Cold bending mandrel maximum diameter (inches)
		Diameters				
	Size Navy Standard	Over insulation minimum (inch)	Over (A) braid covering maximum (inch)	Minimum (inch)	Maximum (inch)	
TCJU-4	4(7)	0.105	0.140	0.400	0.430	6
TCTU-4	4(7)	.105	.140	.400	.430	6

1/ Requirements formerly covered on MIL-C-2194/10.

(A) Denotes changes

FSC 6145

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Voltage withstand - Volts, rms, minimum	3000
Insulation resistance - Megohms/1000 ft., minimum	500
Conductor resistance - Ohms/1000 ft., maximum	
(a) Constantan	71.5
(b) Iron	17.5
(c) Copper	2.51
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Abrasion resistance - Scrapes, minimum	250
Crack resistance	No damage
Group B:	
Gas flame - 1 hour	No failure
(A) Drip - At 95° ± 1°C.	Zero
Cold bending, cable - (See details for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - psi, minimum	750
(A) Elongation - Percent, minimum	125
Jacket	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
(A) Jacket	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (conductor) - Cycles, minimum	50
Permanence of printing (jacket) - Cycles, minimum	250
Cable filler removability	

(A) QUALIFICATION INSPECTION

(A) Qualification inspection shall include basic elelctrical, all of groups A, B and C plus the following:

(A) Cable aging and compatibility (125° ± 5°C.)	No failure
(A) Cold working (minus 20° ± 2°C.)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
TCJU-4 and TCTU-4	2000

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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